

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=0 success=False nfev=80 njev=72

message=The maximum number of function evaluations is exceeded.

reported optimality=0.6993044507030248

```
initial={'cost': 835.7592113256753, 'gradient_inf_norm': 6.603311730816307, 'gradient_l2_norm':
6.925882203902112, 'jacobian': {'condition_number_nonzero_subspace': 565013.5211095744,
'numerical_rank': 13, 'numerical_tolerance': 3.7811920510711364e-10, 'singular_values':
[399.1789759076477, 191.97517693476456, 178.7860312948424, 26.55918317875706,
11.265965727879284, 8.231521320312076, 3.377964602576333, 2.337577928798323,
1.499123824856587, 0.4717623455204067, 0.16264583367886248, 0.07172072580648747,
0.0007064945545440036, 5.2692567520736634e-15]}, 'residual_l2_norm': 40.884207496921725}
final={'cost': 835.7330744320466, 'gradient_inf_norm': 0.6993044507030248, 'gradient_l2_norm':
1.0924033205756185, 'jacobian': {'condition_number_nonzero_subspace': 264480466.89701566,
'numerical_rank': 13, 'numerical_tolerance': 4.0782461811506383e-10, 'singular_values':
[430.538865020046, 193.94705284557844, 178.81617031862078, 28.03845936336991,
11.30015279805276, 8.691807580491597, 3.621528217255149, 2.6027426330780257,
1.5415117330912986, 0.506316927841234, 0.1669136663657599, 0.07089488384678218,
1.627866398117373e-06, 7.073929675980963e-15]}, 'residual_l2_norm': 40.8835682012235}
coordinate_step_l2_norm=3.0791821753654385
cost_reduction=0.026136893628745383
gradient_tolerance_satisfied=False
finite_difference_max_relative_error=3.2830777040459686e-10
finite_difference_directional_checks=[{'analytic_l2_norm': 296.21987927803974,
'difference_l2_norm': 9.725128814550341e-08, 'direction': 'negative gradient',
'finite_difference_l2_norm': 296.21987937006116, 'relative_error': 3.2830777040459686e-10,
'scheme': 'central', 'step': 6.642216157943094e-05}]
```

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.